

Theo Usher

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EDUCATION

Columbia University, *Bachelor of Science, Mechanical Engineering* New York, NY; Sep 2020 – May 2024
GPA 4.07/4.0, Tau Beta Pi Engineering Honor Society, Mechanical Engineering Certificate of Merit, Magna Cum Laude
Electives: Robotics Studio, Managing Tech Innovation, Public Speaking, Human-Centered Design

WORK EXPERIENCE

Boeing, *Space Electronics and Mechanical Product Design Engineer* Los Angeles, CA; Aug 2024 – Present

- Serve as Mechanical Responsible Engineering Authority for Spacecraft Control Electronics
- Own electromechanical hardware design from concept through prototyping, testing, and production
- Create detailed CAD models, drawings, and tolerance analyses with GD&T for electromechanical assemblies
- Design multilayer PCBs with high-speed signal routing and thousands of densely packed components
- Integrate PCBAs, wire harnesses, and electromechanical components into tightly packaged mechanical housings
- Design mechanical systems balancing load paths, tolerances, thermals, vibration, and manufacturing constraints
- Conduct vibration testing and failure analysis of flight units, instrumenting to characterize limits
- Lead production support, coordinating manufacturing teams to disposition millions in flight hardware

Airobotics Drones, *Design Engineering Intern* Tel Aviv, Israel; May 2023 – Jul 2023

- Designed mechanical subsystems for the Iron Drone, an autonomous counter-drone
- Analyzed, tested, and iterated on the drone's net launcher for range, reliable interception, and safe capture
- Redesigned the drone ground station for improved usability, reliability, and performance
- Prototyped and optimized the drone ground station, cutting assembly time from 2 weeks to 2 days
- Engineered the docking interface governing how the drone mates with its ground station
- Delivered engineering used in Iron Drone systems deployed globally and exhibited at the U.S. DOT

Terabase Energy, *Mechanical Engineering Intern* Davis, CA; May 2022 – Aug 2022

- Designed and field-tested autonomous robotic vehicles that handle and install solar panels at utility scale
- Engineered a solar panel-handling end effector that optimized load paths and doubled installation range
- Built a panel stabilization system that replaced high-cost linear actuators with low-cost, durable mechanisms
- Prototyped a user-friendly solar panel rack for rapid field deployment, taking it from concept to trials in weeks

ENGINEERING PROJECTS & LEADERSHIP

Columbia Space Initiative, *Rockets Mission Lead* Sep 2020 – Jun 2024

- Led a 50-person team to research, design, and build a hybrid rocket to reach 30,000 ft with a scientific payload
- Managed cross-subsystem integration for rocket superstructure, mechanisms, sensors, and wiring
- Led manufacturing across a range of processes including CNC machining, producing all of our rockets in-house
- Used structural FEA and hand calcs to identify and correct critical failure points in the rocket superstructure
- Created safe test procedures, including static fires and recovery tests, to optimize structures and components
- Achieved Columbia's first successful hybrid flight and recovery, improved engine performance by 10x
- Designed, built, and launched a new and improved rocket every year, a rare cadence for a student team

Automated Robotic Linkage – Machine Design Sep 2023 – Dec 2023

- Designed, built, and programmed a complex linkage and closed-loop control to rapidly press arcade buttons
- Built a cable-driven power transmission unlike any our professor had seen, greatly reducing inertia

Bipedal Walking Robot Jan 2022 – May 2022

- Designed a servo-driven bipedal robot and programmed its control software from scratch
- Built the first and fastest walking robot in a class of students several years ahead of me

SKILLS

Technical Skills: CAD (SolidWorks, Creo), Robotics, Mechanism Design, End Effectors, Actuators, PCB Design, Circuit Design, Wire Harnessing, Controls, FEA, GD&T, CNC Machining, Rapid Prototyping, MATLAB, Python

Non-Technical Skills: Project Management, Cross-Functional Collaboration, Public Speaking, Documentation